

IPL CRUNCH '26 : IPL Data Analysis Using Python and Pandas

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ABSTRACT

The Indian Premier League (IPL) generates a large amount of ball-by-ball cricket data every season. This data can be analyzed to identify performance trends, player impact, and match patterns.

The aim of this project was to analyze IPL data using Python and Pandas and interpret the results with the help of statistical analysis and visualizations. The analysis focused on batting and bowling performances, strike-rate-based comparisons, and match outcomes.

The dataset used for this project was matches.csv, which contains detailed ball-by-ball records of IPL matches. The primary tools used were Python, Pandas, and Matplotlib.

Through this project, key insights were identified such as top run scorers, leading wicket takers, comparisons between slow anchor innings and quick cameo innings, and patterns affecting team performance. The project also demonstrates how data science techniques can be applied effectively in sports analytics.

1. INTRODUCTION

1.1 IPL Overview

The Indian Premier League (IPL) is one of the most popular T20 cricket tournaments in the world. It is held annually in India and features franchise teams representing different cities. Players from across the world, including international, capped, and uncapped players, participate in the tournament and showcase their skills.

Strong performances in the IPL often create opportunities for players at both national and international levels. Since every delivery in cricket contributes to match statistics, IPL produces a large volume of valuable data. Cricket itself is a strategic and statistical sport, and this data plays an important role in identifying patterns, comparing performances, and supporting future decision-making.

1.2 Why Analyze IPL Data?

Analyzing IPL data helps in understanding player and team performance more effectively. It is useful for :

- Evaluating batting and bowling performances
- Comparing players using statistical measures
- Identifying trends and patterns in matches
- Supporting strategic planning for future games
- Understanding how data science can be applied in sports analytics

IPL statistics are also widely used in match broadcasts, commentary, and performance analysis.

1.3 Project Objective

The objective of this project is to analyze the IPL dataset and answer important performance-related questions such as:

- Do teams that win the toss also win more matches?
- Which match phase impacts victory the most - Powerplay, Middle Overs, or Death Overs?
- Who are the top-performing batters and bowlers across seasons?
- What hidden patterns or surprising trends can be identified from the dataset?

2. DATASET DESCRIPTION

- Dataset : matches.csv
- The dataset contains detailed ball-by-ball records of IPL matches. Each row represents one delivery bowled in a match.
- Columns :
 - match_id
 - date
 - season
 - event
 - venue
 - city
 - team1
 - team2
 - toss_winner
 - toss_decision
 - winner
 - win_by_runs
 - win_by_wickets
 - player_of_match
 - innings
 - batting_team
 - over
 - ball
 - batter
 - bowler
 - non_striker
 - runs_batter
 - runs_extras
 - runs_total
 - extras_wides
 - extras_noballs
 - extras_byes
 - extras_legbyes
 - wicket_kind
 - wicket_player_out
- No of rows = 289673
- No of columns = 30

3. TOOLS AND TECHNOLOGIES

The following tools were used in this project:

- Python - Programming language used for analysis
- Pandas - Data loading, cleaning, grouping, and analysis
- Matplotlib - Data visualization and chart creation
- VS Code - Integrated Development Environment (IDE)

4. METHODOLOGY

The project was completed through a structured data analysis process using Python and Pandas.

4.1 Data Loading

The IPL dataset was imported into Python using Pandas.

Steps performed:

- Loaded the dataset from matches.csv
- Verified the dataset structure
- Checked column names
- Examined the number of rows and columns
- Reviewed sample records for understanding the data

4.2 Data Preprocessing

Before analysis, the dataset was cleaned and organized.

The preprocessing steps included:

- Checking for missing or null values
- Verifying data consistency
- Grouping records by batter and bowler
- Filtering wicket-related deliveries
- Creating strike-rate calculations for batters
- Preparing summarized tables for analysis

This step ensured that the dataset was ready for accurate analysis.

4.3 Data Analysis Performed

Different analyses were carried out to answer the project objectives. These included:

Batting Analysis

- Total runs scored by each batter
- Top 5 batters by total runs
- Strike-rate calculations
- Slow anchor innings analysis
- (50+ runs with strike rate below 150)
- Quick cameo innings analysis
- (25+ runs with strike rate above 200)

Bowling Analysis

- Total wickets taken by bowlers
- Top 5 bowlers by wickets

Match Outcome Analysis

- Toss winner vs match winner comparison
- Phase-based impact on results
- Identification of patterns and unexpected outcomes

4.4 Data Visualization

Matplotlib was used to visualize the results. Visualizations created included:

- Bar charts for top batters
- Bar charts for top bowlers
- Comparison charts for slow anchors vs quick cameos
- Summary tables for batting and bowling performance

These visualizations made it easier to identify trends and interpret the findings clearly.

5. ANALYSIS AND RESULT

Q.1 Do teams that win the toss actually win more matches?

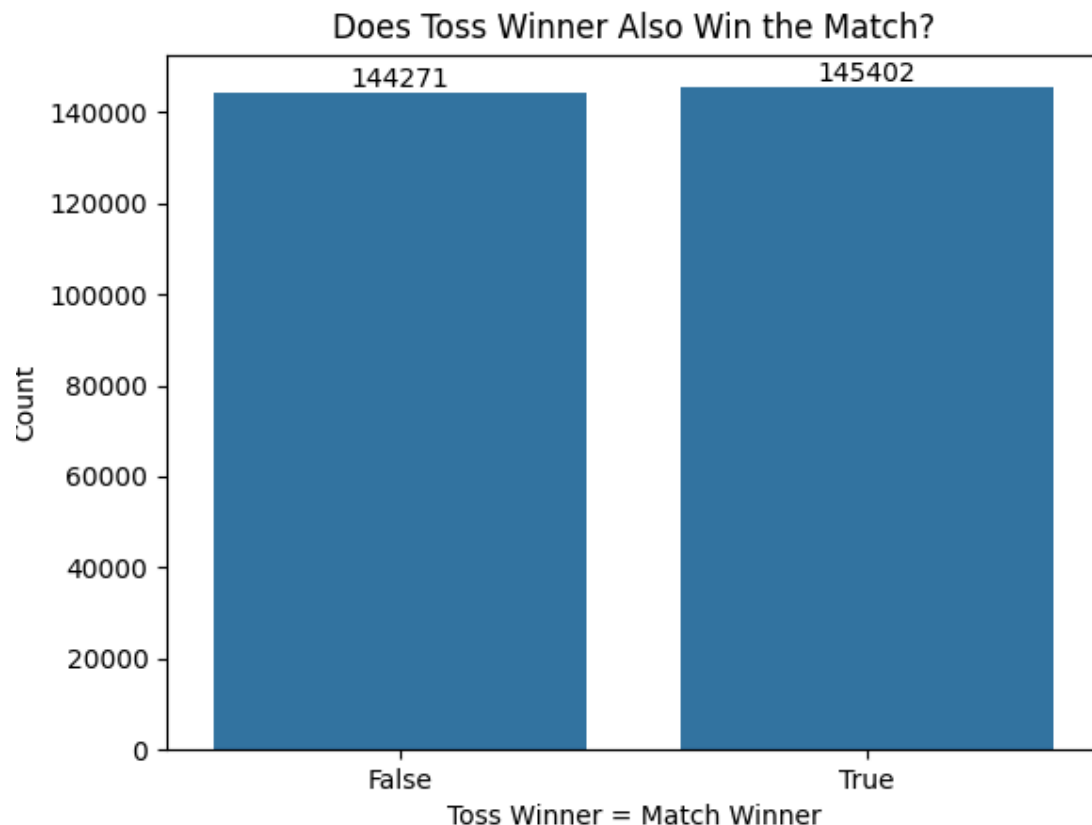
One of the objectives of this project was to determine whether winning the toss gives teams a clear advantage in winning IPL matches.

To analyze this, a new column called `toss_match_win` was created. This column compared the values in `toss_winner` and `winner`.

- If both values were the same, the result was marked as True
- If the values were different, the result was marked as False

Result

Toss winner also won match	Count
True	1,45,402
False	1,44,271



Winning Percentage

Winning percentage = 50.20%

Therefore:

- Teams that won the toss and also won the match: 50.20%
- Teams that won the toss but lost the match: 49.80%

Interpretation

The results show that teams winning the toss also won the match in 50.20% of cases. This percentage is only slightly higher than teams that lost despite winning the toss.

This indicates that winning the toss does not provide a major advantage in IPL matches. While toss decisions such as batting or bowling first may influence strategy depending on pitch and match conditions, the overall impact on the final result is very small.

The difference between winning and losing after the toss is almost equal, which suggests that match outcome depends more on factors such as:

- Team performance
- Batting partnerships
- Bowling effectiveness
- Fielding quality
- Match pressure and decision-making

Conclusion

Based on the IPL dataset analysis, it can be concluded that winning the toss alone does not significantly increase the chances of winning the match.

Although toss decisions may offer a slight tactical advantage in certain situations, the final outcome is mainly determined by overall team performance during the match.

Q.2 Which phase impacts victory the most - Powerplay, Middle Overs, or Death Overs?

To identify which phase of a T20 match has the greatest impact on victory, the match was divided into three phases:

- Powerplay: Overs 1–6
- Middle Overs: Overs 7–15
- Death Overs: Overs 16–20

A phase impact scoring method was used.

For each match:

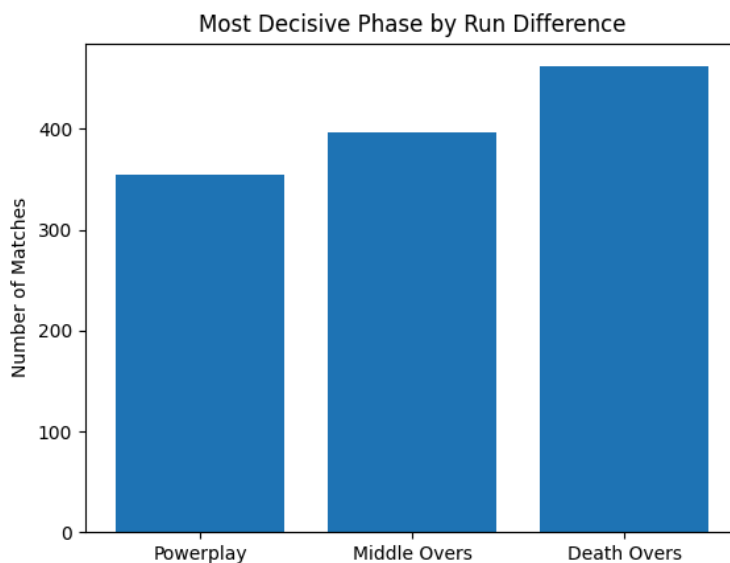
1. The difference in runs scored by both teams in each phase was calculated.
2. The phase with the highest run difference received +1 impact score.
3. The difference in wickets taken by both teams in each phase was calculated.
4. The phase with the highest wicket difference received +1 impact score.

This process was repeated for all matches in the dataset.

The phase with the highest total score was considered the most influential phase.

Run Difference Impact Scores

Phase	Impact Score
Powerplay	354
Middle Overs	396
Death	462



Interpretation

Among run-based comparisons, the Death Overs had the highest impact score.

This shows that in most IPL matches, the largest scoring difference between teams was created during the final overs.

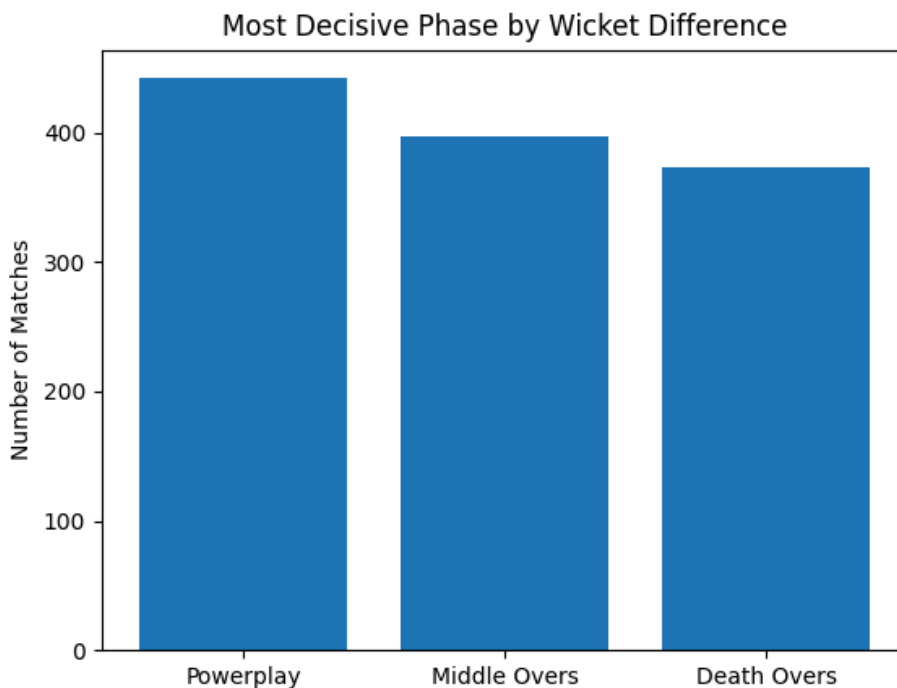
This indicates:

- Teams often accelerate aggressively in the last overs
- Finishing ability contributes heavily to the total score
- A strong death-over performance can significantly shift match momentum

Middle overs also had an important contribution, while Powerplay had the lowest run-based impact.

Wicket Difference Impact Scores

Phase	Impact Score
Powerplay	442
Middle Overs	397
Death	373



Interpretation

For wicket-based comparisons, the Powerplay recorded the highest impact score.

This indicates that the biggest wicket differences happened most frequently in the opening overs.

This suggests:

- Early wickets put pressure on batting teams
- Losing top-order batters affects innings stability
- Strong Powerplay bowling creates an early advantage

Middle overs and death overs also influenced results, but less frequently than the Powerplay.

Overall Conclusion

The analysis shows that different phases influence matches in different ways.

- **Powerplay was the most decisive phase for wickets**
- **Death Overs were the most decisive phase for runs**

This suggests that IPL matches are often shaped by:

- **Early breakthroughs during Powerplay overs**
- **High scoring and finishing strength during Death Overs**

Therefore, both the beginning and ending phases of a T20 innings play a crucial role in determining match outcomes, while middle overs mainly help teams maintain momentum between these two key phases.

To identify the most decisive phase in IPL matches, run-difference scores and wicket-difference scores were combined.

Phase	Run Impact	Wicket Impact	Total Impact
Powerplay	354	442	796
Middle Overs	396	397	793
Death Overs	462	373	835

After combining both batting and bowling impact scores, Death Overs recorded the highest overall score **(835)**.

The reason is that the death overs often involve:

- aggressive scoring from batters
- quick momentum shifts
- pressure on bowlers
- crucial wickets in the final stage of the innings

Although Powerplay had the highest wicket impact, the strong run impact in Death Overs made it the most decisive phase overall.

Middle overs also contributed consistently, but had slightly lower total impact.

Based on the analysis, **Death Overs were the most determining phase in IPL matches.**

This suggests that finishing strongly in the final overs—through fast scoring or taking important wickets—has the greatest influence on deciding the outcome of a match.

Q.3 Who are the top batters and bowlers across seasons?

To identify the top-performing players across IPL seasons, batting and bowling statistics were calculated from the ball-by-ball dataset.

For batters :

- Total runs scored were calculated using runs_batter
- Total balls faced were counted
- Strike rate was calculated using: $\text{strike_rate} = (\text{runs}/\text{ball}) * 100$

For bowlers:

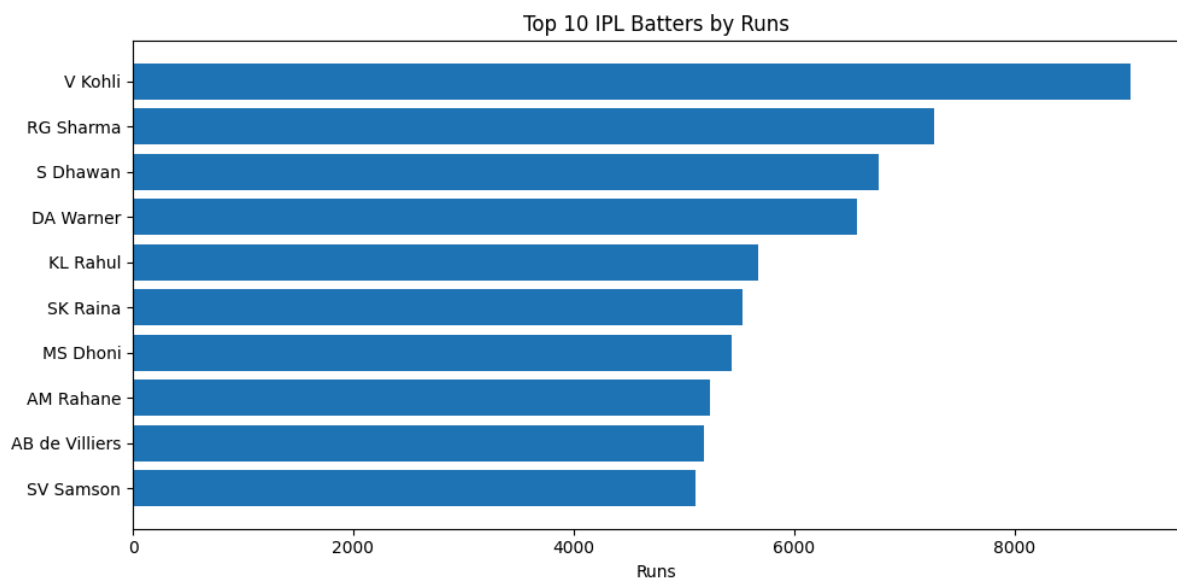
- Only valid wickets were considered:
 - bowled
 - caught
 - lbw
 - caught and bowled
 - stumped
 - hit wicket
- Total wickets were counted
- Runs conceded were calculated
- Economy rate was calculated using: $\text{economy} = (\text{runs_conceded}/\text{balls_bowled}) * 6$

The top 10 players were selected based on total runs and wickets.

Top 10 batters across all seasons by runs

Batter	Runs	Balls	Strike Rate
V Kohli	9050	6942	130.365889
RG Sharma	7269	5605	129.687779
S Dhawan	6769	5483	123.454313
DA Warner	6567	4849	135.429986
KL Rahul	5680	4205	135.077289
SK Raina	5536	4177	132.535312

MS Dhoni	5439	4101	132.626189
AM Rahane	5237	4296	121.904097
AB de Villiers	5181	3487	148.580442
SV Samson	5106	3721	137.221177



Analysis

Among all batters, **Virat Kohli** recorded the highest number of runs with **9050** runs, making him the leading run scorer in the dataset.

Rohit Sharma and **Shikhar Dhawan** followed closely, showing strong long-term consistency across multiple seasons.

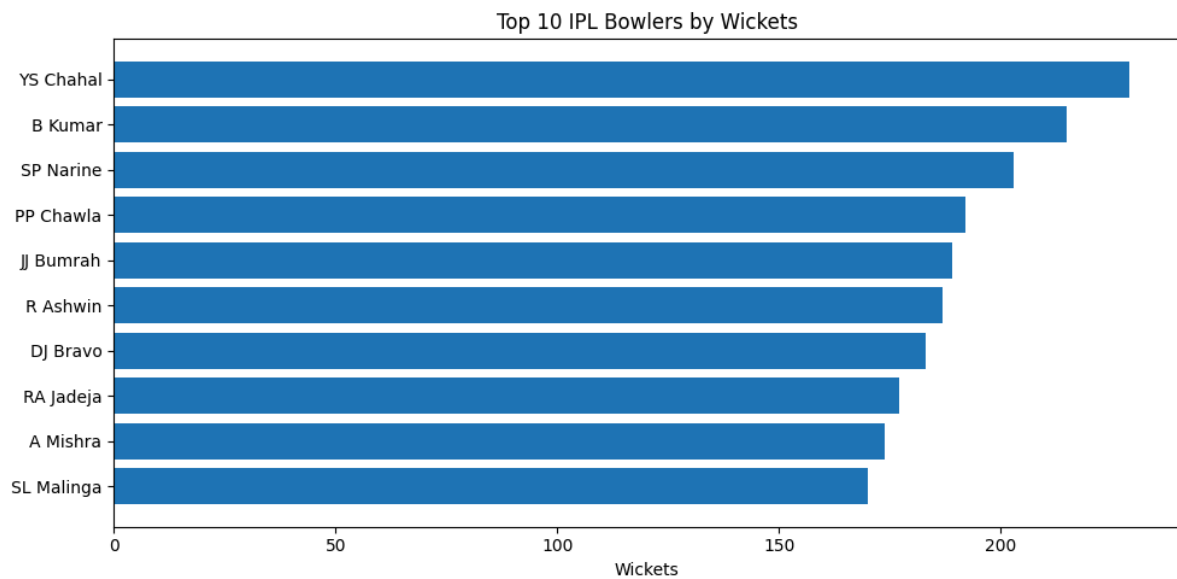
A notable observation was **AB de Villiers**, who maintained the highest strike rate (**148.58**) among the top run scorers. This shows that he scored at a faster pace while also maintaining a high run tally.

The batting data suggests that top IPL batters combine:

- consistency across seasons
- ability to score under pressure
- strong strike-rate management in T20 cricket

Top 10 bowlers across all seasons by wickets

Bowler	Wickets	Economy
YS Chahal	229	7.907294
B Kumar	215	7.582245
SP Narine	203	6.825163
PP Chawla	192	7.977920
JJ Bumrah	189	7.262988
R Ashwin	187	7.051356
DJ Bravo	183	8.075243
RA Jadeja	177	7.629195
A Mishra	174	7.304878
SL Malinga	170	7.032952



Analysis

Among bowlers, Yuzvendra Chahal recorded the highest number of wickets with 229 wickets.

Bhuvneshwar Kumar and Sunil Narine were also among the most successful wicket takers.

A notable observation is Sunil Narine, who maintained the best economy rate (6.83) among the top wicket takers, indicating excellent control while also taking wickets regularly.

This analysis shows that successful IPL bowlers generally combine:

- wicket-taking ability
- maintaining pressure through lower economy
- consistency across multiple seasons

Conclusion

The analysis highlights the players who have consistently performed at the highest level across IPL seasons.

Key findings:

- **Virat Kohli was the leading run scorer**
- **AB de Villiers had the highest strike rate among top batters**
- **Yuzvendra Chahal was the leading wicket taker**
- **Sunil Narine maintained the most economical bowling among the top wicket takers**

These results show that long-term success in IPL depends on both consistency and impact, whether through scoring quickly with the bat or taking wickets while controlling runs with the ball.

Q.4 What hidden patterns or surprises can you discover from the data?

1. Strike Rate on Free Hit Deliveries

One of the most interesting patterns identified from the IPL dataset was the outcome of free hit deliveries.

A free hit is generally viewed as a high-scoring opportunity because the batter cannot be dismissed in the usual ways. Viewers often expect aggressive batting, boundaries, and very high scoring on these deliveries.

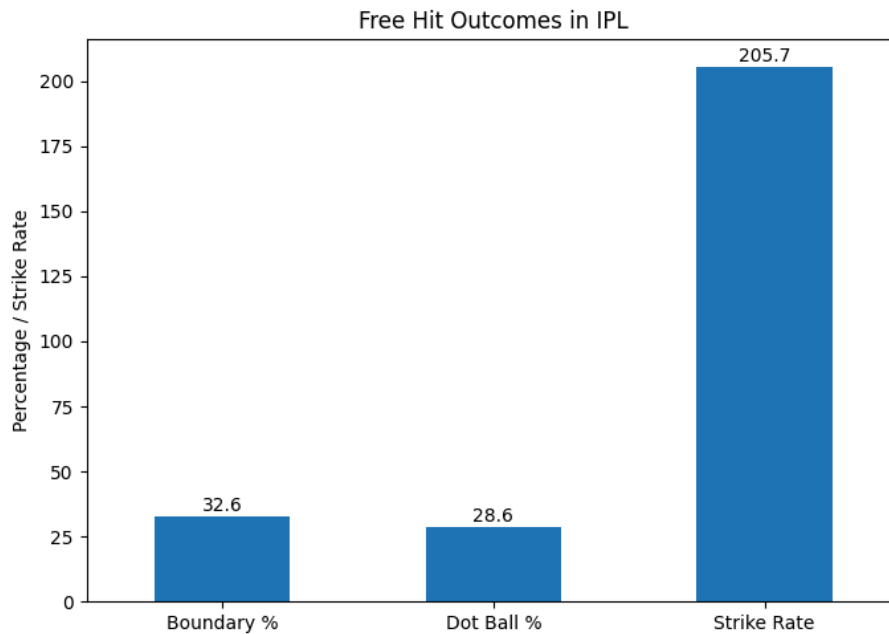
To analyze this, all deliveries immediately following a no-ball were identified as free-hit balls.

The following metrics were calculated:

- Total free-hit deliveries
- Strike rate
- Boundary percentage
- Dot-ball percentage
- Average runs per delivery

Result

Metric	Value
Free-hit deliveries	1,198
Strike rate	205.68
Boundary percentage	32.55
Dot-ball percentage	28.63
Average runs	2.06



Analysis

At first glance, many viewers may expect free-hit deliveries to produce extremely high scoring - possibly a boundary almost every time - because the batter can attack without the normal risk of getting out.

However, the actual numbers reveal a different pattern.

Although the strike rate on free-hit balls was high at 205.68, it was lower than many would expect.

A particularly surprising finding was that:

- only 32.55% of free-hit deliveries resulted in boundaries
- nearly 28.63% were still dot balls

This means that almost one out of every three free-hit deliveries produced no runs at all, despite the batter having freedom to attack.

Possible reasons include:

- bowlers using slower balls or yorkers to reduce scoring
- batters attempting aggressive shots and mis-timing them
- pressure situations leading to risky decision-making
- field placements restricting easy boundaries

Conclusion

The analysis shows that free-hit deliveries are valuable scoring opportunities, but they are not as dominant as viewers often assume.

Even without the normal dismissal pressure:

- **boundaries are not guaranteed**
- **dot balls still occur frequently**
- **the average scoring remains just over 2 runs per ball**

This highlights an interesting aspect of IPL cricket: even when the batter has an advantage, bowling strategy and execution can still limit scoring effectively.

The result demonstrates that expectations around free hits are often greater than what actually happens on the field.

2. Slow Anchors VS Quick Cameos

Another interesting pattern identified from the IPL dataset was comparing slow anchor innings and quick cameo innings.

In T20 cricket, it is often assumed that a batter scoring a big 50 or more automatically contributes more to victory than a batter scoring fewer runs. A longer innings also looks more valuable because it helps stabilize the batting order and often leads to personal milestones.

However, the IPL data showed a surprising result.

For this analysis:

Slow Anchor innings were defined as:

- 50 or more runs
- Strike rate below 130

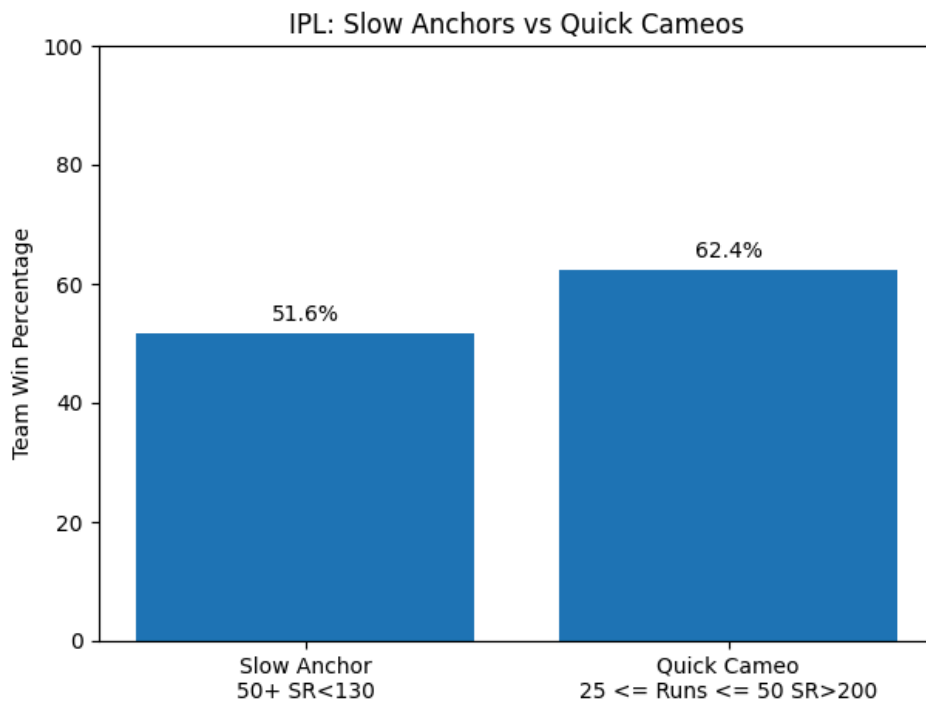
Quick Cameo innings were defined as:

- 25 to 50 runs
- Strike rate above 200

The winning team for each match was then compared with the batting team of these innings.

Result

Innings Type	Total Innings	Total Win Percentage
Slow Anchor (50+ runs, SR <130)	397	51.6%
Quick Cameo (25–50 runs, SR >200)	354	62.4%



Analysis

The results showed that quick cameo innings had a noticeably stronger connection with winning matches than slow anchor innings.

Even though slow anchors scored more runs and stayed at the crease longer, teams with these innings won only 51.6% of the time.

In comparison, teams featuring quick cameo innings won 62.4% of the time.

This is surprising because:

- slow anchors produce larger individual scores
- they often appear more dependable
- they contribute toward milestones like half-centuries

Yet in the T20 format, momentum and scoring speed often matter more than total runs alone.

A quick cameo can:

- rapidly increase the run rate
- shift pressure onto bowlers
- create momentum in a few overs
- finish innings strongly during crucial phases

A slow anchor may still be useful, but if the scoring rate remains low, it can reduce momentum and increase pressure on the remaining batters.

Conclusion

The analysis suggests that in IPL T20 cricket, quick explosive batting can be more impactful than longer but slower innings.

Although slow anchor innings may look stronger on paper because of the higher run total, the data indicates that aggressive cameos are more strongly associated with match wins.

This highlights an important T20 insight : **In short-format cricket, how quickly runs are scored can matter more than how many runs are scored individually.**

6. EXTRA CHARTS

Chart 1 : two bars showing win rate of toss winners vs toss losers

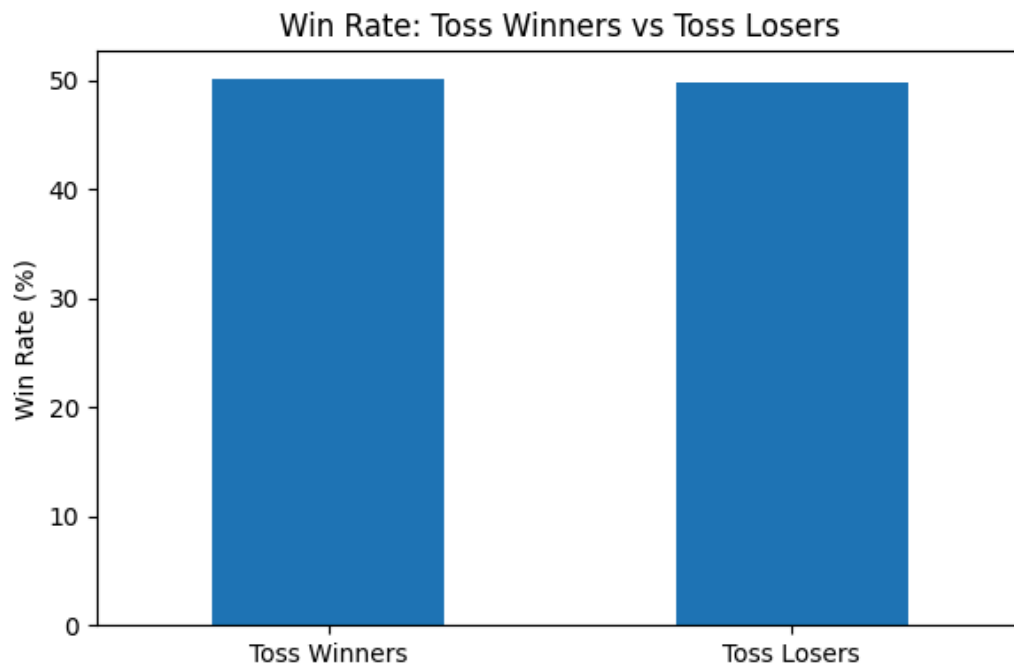
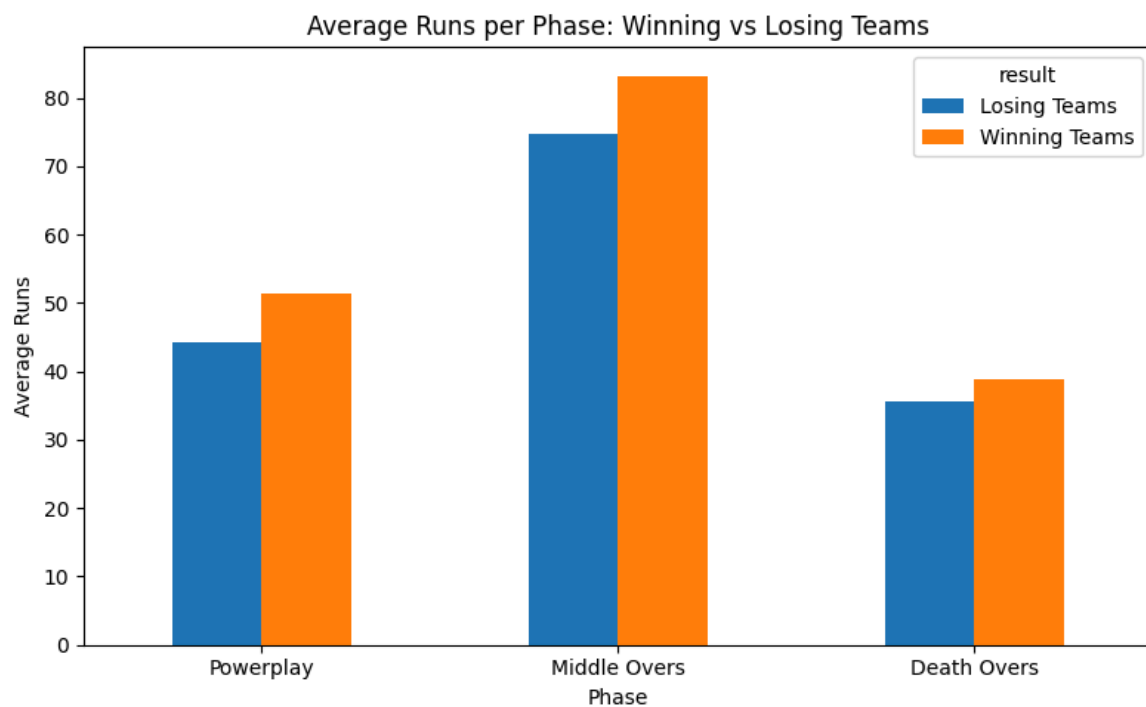


Chart 2 : average runs per phase for winning teams vs losing teams



7. CONCLUSION

This project demonstrated how Python and Pandas can be effectively used to analyze large sports datasets and extract meaningful insights.

Working with nearly 2.9 lakh IPL ball-by-ball records was made highly efficient using Pandas, which made data cleaning, grouping, filtering, and aggregation efficient. Matplotlib helped convert the numbers into visual charts and graphs, making patterns easier to understand and compare.

The analysis revealed several useful insights from IPL data:

- Teams winning the toss had only a slight advantage, showing that toss alone does not strongly determine match results.
- Different match phases impact outcomes differently, with later overs often becoming crucial due to faster scoring and higher pressure.
- Batting strike rate plays a major role in T20 cricket, often influencing outcomes more strongly than total runs scored.
- Top batters and bowlers consistently maintained performance over multiple seasons.
- Hidden patterns such as free-hit scoring and quick cameos revealed that common assumptions in cricket are not always supported by data.

Overall, the project highlighted how data analysis can uncover patterns that may not be obvious while watching matches. It also showed the practical application of Python in sports analytics and how statistical interpretation can support cricket strategy and performance evaluation.

8. FUTURE SCOPE

This project can be extended further in several ways.

Possible future improvements include:

Match Winner Prediction

Machine learning models can be built using historical IPL data to predict match outcomes based on factors such as venue, toss, innings score, wickets, and player performance.

Player Performance Forecasting

Future player performance can be estimated using previous seasons' data, including batting form, strike rate, wickets, and consistency.

Team Strategy Recommendations

The analysis can be expanded to suggest team strategies such as:

- ideal batting order
- best death-over bowlers
- powerplay scoring strategies
- player matchups against specific opponents
- Advanced Visualization Dashboards

Interactive dashboards can be created using tools like Plotly or Streamlit for easier exploration of player and team statistics.

Machine Learning Applications

The dataset can be used for:

- score prediction
- win probability prediction
- player impact analysis
- identifying high-pressure performers

These additions can make the project more advanced and bring it closer to real-world sports analytics systems used by teams and broadcasters

9. REFERENCES

1. IPL Ball-by-Ball Dataset - Kaggle / provided project dataset matches.csv
2. Python Documentation - <https://docs.python.org/3/>
3. Pandas Documentation - <https://pandas.pydata.org/docs/>
4. Matplotlib Documentation - <https://matplotlib.org/stable/>
5. Seaborn Documentation - <https://seaborn.pydata.org/>
6. Visual Studio Code Documentation - <https://code.visualstudio.com/docs>